ICT 6535: Advanced Artificial Intelligence

Lecture 2 — Agents in AI

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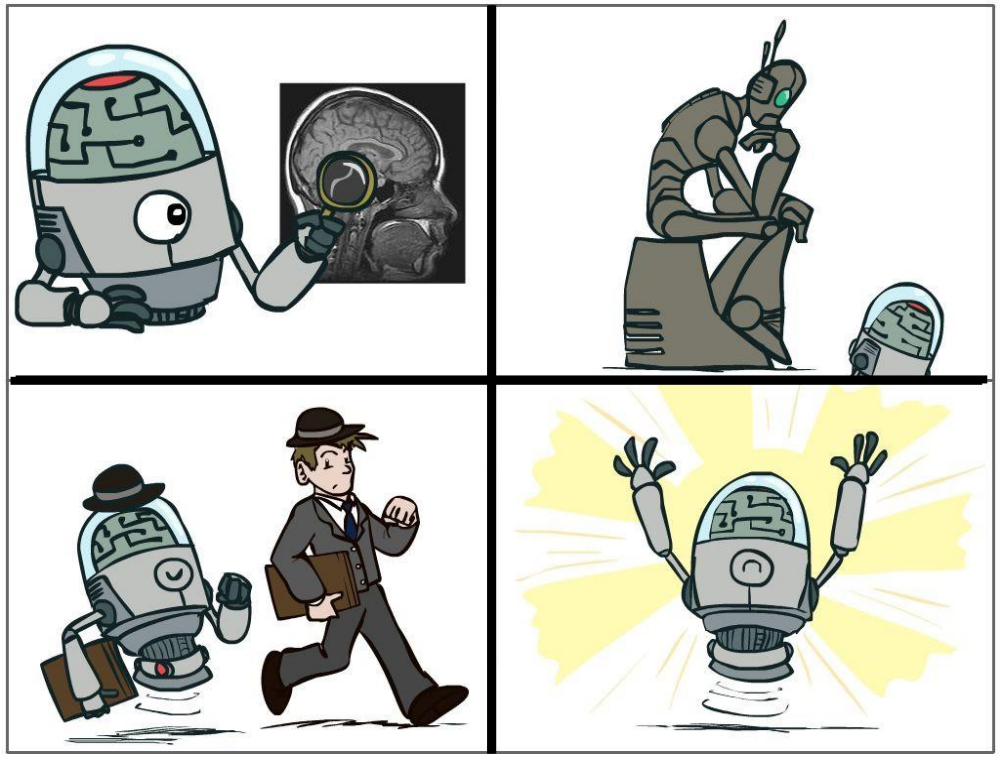
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Artificial Intelligence

Artificial Intelligence is the science of making machines that

Think like people



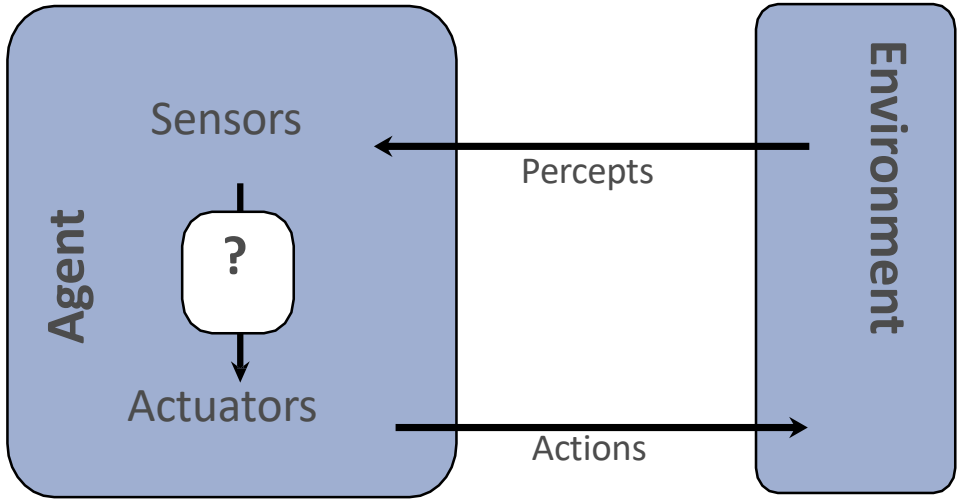
Think rationally

Act like people

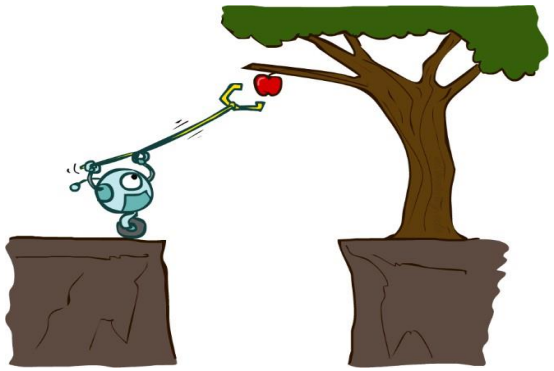
Act rationally

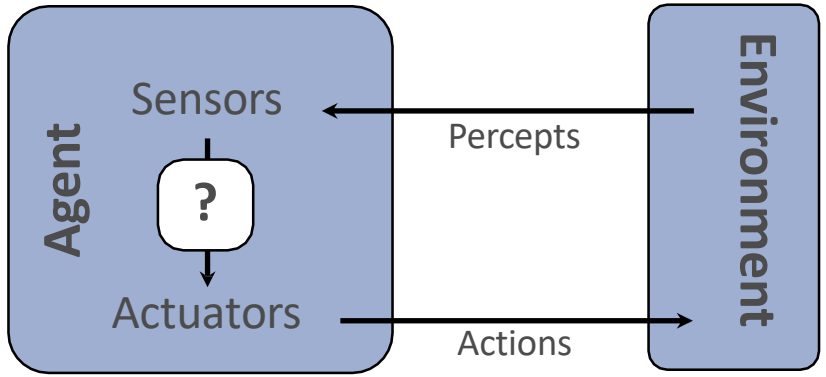
Agents in Artificial Intelligence

- Artificial intelligence is defined as the study of rational agents.
- An agent could be anything that makes decisions, as a person, firm, machine, or software.
- An agent is anything that can be viewed as
  - perceiving its environment through **sensors** and
  - acting upon that environment through **actuators**



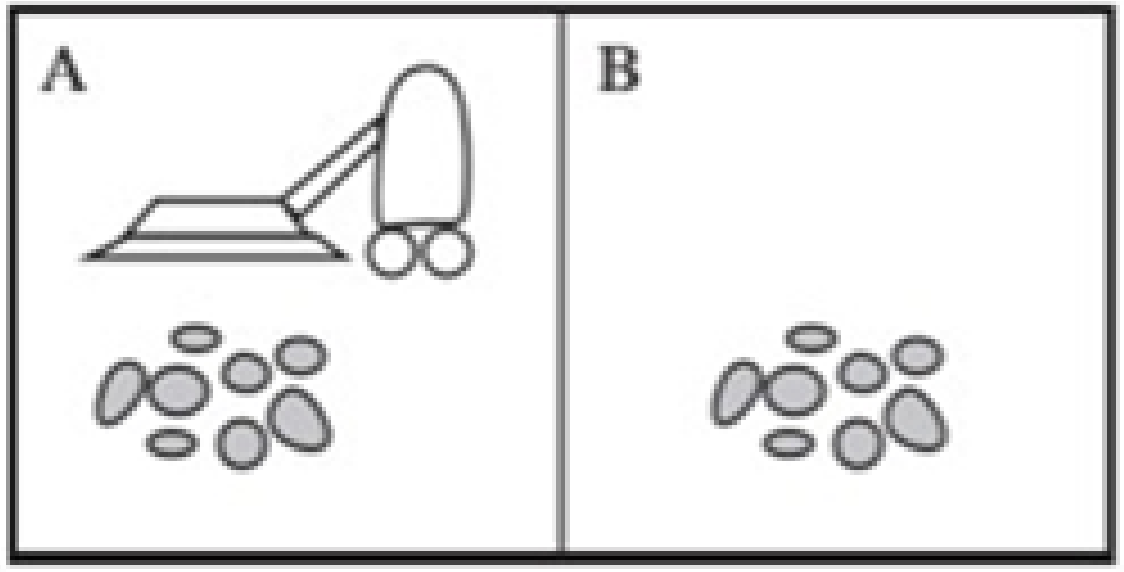
Example of Agents





- A Human-agent: eyes, ears and other organs which act as sensors, and hands, legs, mouth, and other body parts acting as actuators.
- A Robotic agent: Cameras and infrared range finders which act as sensors and various motors acting as actuators.
- A software agent: Keystrokes, file contents, received network packages which act as sensors and displays on the screen, files, sent network packets acting as actuators.

Example of Agents: A Vacuum Cleaner



Sensors?

– Camera, other sensors

Actuators?

– Wheels, robotic arms etc.

- Task: If there is a dirt, clean (suck) it, otherwise move to the other room.

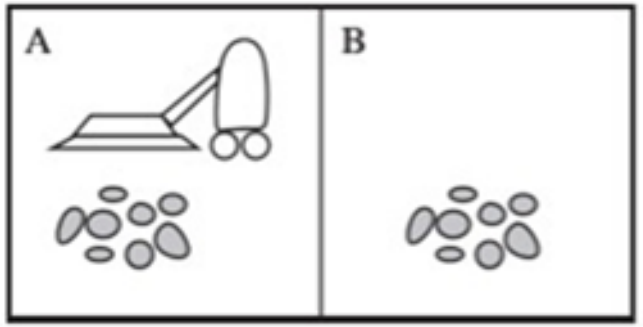
Agents in Artificial Intelligence

- The term *percept* refers to the agent's perceptual inputs at any given instant.
- An agent's *percept sequence* is the complete history of everything the agent has ever perceived.
- In general, an agent's choice of action at any given instant can depend on the entire percept sequence observed to date, but not on anything it hasn't perceived.
- Mathematically speaking, we say that an agent's behavior is described by the *agent function* that maps any given percept sequence to an action.
- Internally, the agent function for an artificial agent will be implemented by an *agent program*.

Example of Agents: A Vacuum Cleaner

An example of a partial tabulation of an agent function for the vacuum-cleaner agent

Percept sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck
[A, Clean], [A, Clean]	Right
[A, Clean], [A, Dirty]	Suck
⋮	⋮
[A, Clean], [A, Clean], [A, Clean]	Right
[A, Clean], [A, Clean], [A, Dirty]	Suck
⋮	⋮



Agents in Artificial Intelligence

- An AI system is composed of an agent and its environment.
- An agent needs an architecture and a program to execute the agent function

  Agent = Architecture + Agent Program

- Architecture is the machinery that the agent executes on.
  - It is a device with sensors and actuators, for example, a robotic car, a camera, a PC.
- Agent program is an implementation of an agent function.
  - An agent function is a map from the percept sequence (history of all that an agent has perceived to date) to an action.

Rational Agents in Artificial Intelligence

- **A rational agent is one that does the right thing.**
- **What does it mean to do the right thing?**
  - by considering the consequences of the agent's behavior.
- **When an agent is plunked down in an environment:**
  - it generates a sequence of actions according to the percepts it receives.
  - this sequence of actions causes the environment to go through a sequence of states.
  - if the sequence is desirable, then the agent has performed well.
- **This notion of desirability is captured by a performance measure, evaluating any given sequence of environment states.**

Rationality

- **What is rational at any given time depends on four things:**
  - The performance measure that defines the criterion of success.
  - The agent's prior knowledge of the environment.
  - The actions that the agent can perform.
  - The agent's percept sequence to date.
- **This leads to a definition of a *Rational Agent***
  - For each possible percept sequence, a *rational agent* should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

Task Environment

- Task Environment is essentially the “problems” to which rational agents are the “solutions.”
- In the discussion of the rationality of the simple vacuum-cleaner agent, we had to specify the performance measure, the environment, and the agent’s actuators and sensors. All of these four factors are grouped under the heading of the *Task Environment*.
- PEAS description of a *Task Environment* refers to
  - Performance
  - Environment
  - Actuators
  - Sensors
- For designing an agent, the first step must always be to specify the task environment as fully as possible.

Task Environment for an Automated Taxi Driver

PEAS description of the task environment for an automated taxi

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

Task Environment: More Examples

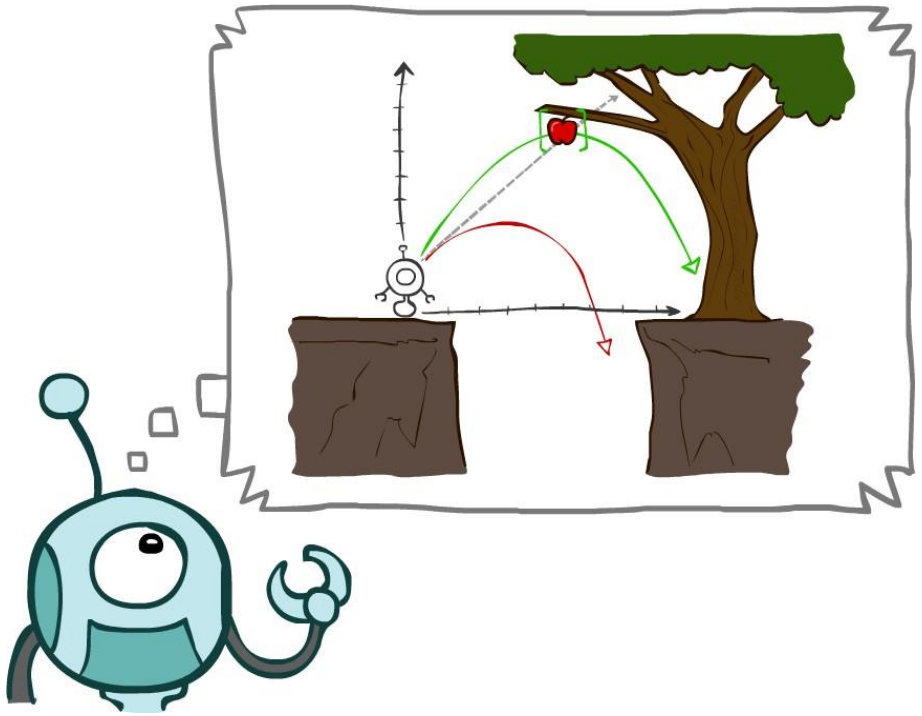
Agent Type	Performance Measure	Environment	Actuators	Sensors
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry

Properties of Environment

- **Environment can be distinguished based on six dimensions:**
  - Fully observable vs. Partially observable
  - Single agent vs. Multiagent
  - Deterministic vs. Stochastic
  - Episodic vs. Sequential
  - Static vs. Dynamic
  - Discrete vs. Continuous

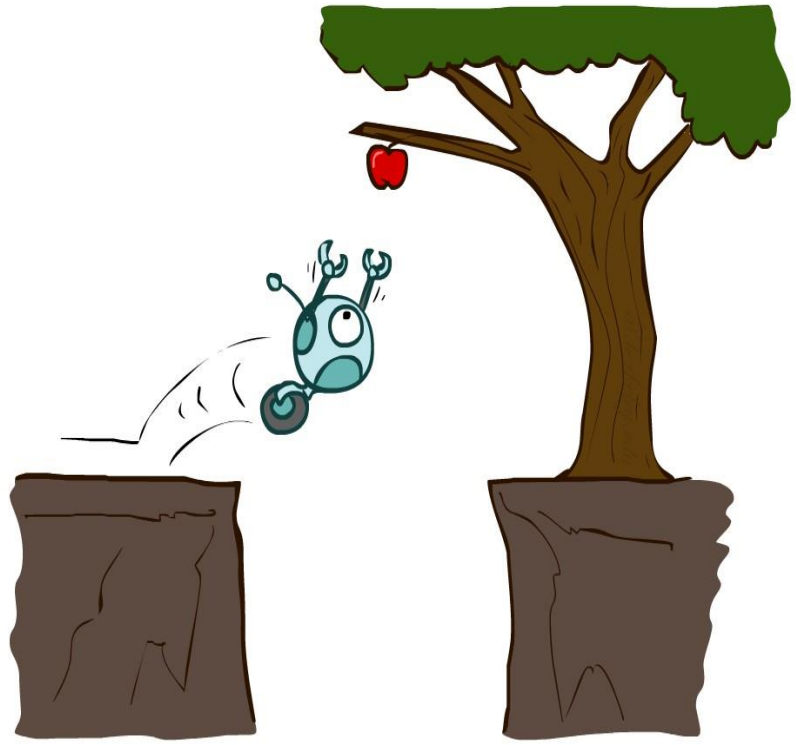
Types of Agents

- **Agents can be grouped into five classes based on their degree of perceived intelligence and capability:**
  - Simple Reflex Agents
  - Model-Based Reflex Agents
  - Goal-Based Agents
  - Utility-Based Agents
  - Learning Agent



Simple Reflex Agents

- **Reflex agents:**
  - Simple reflex agents ignore the rest of the percept history and choose action based on current percept.
  - Do not consider the future consequences of their actions
  - Consider how the world IS

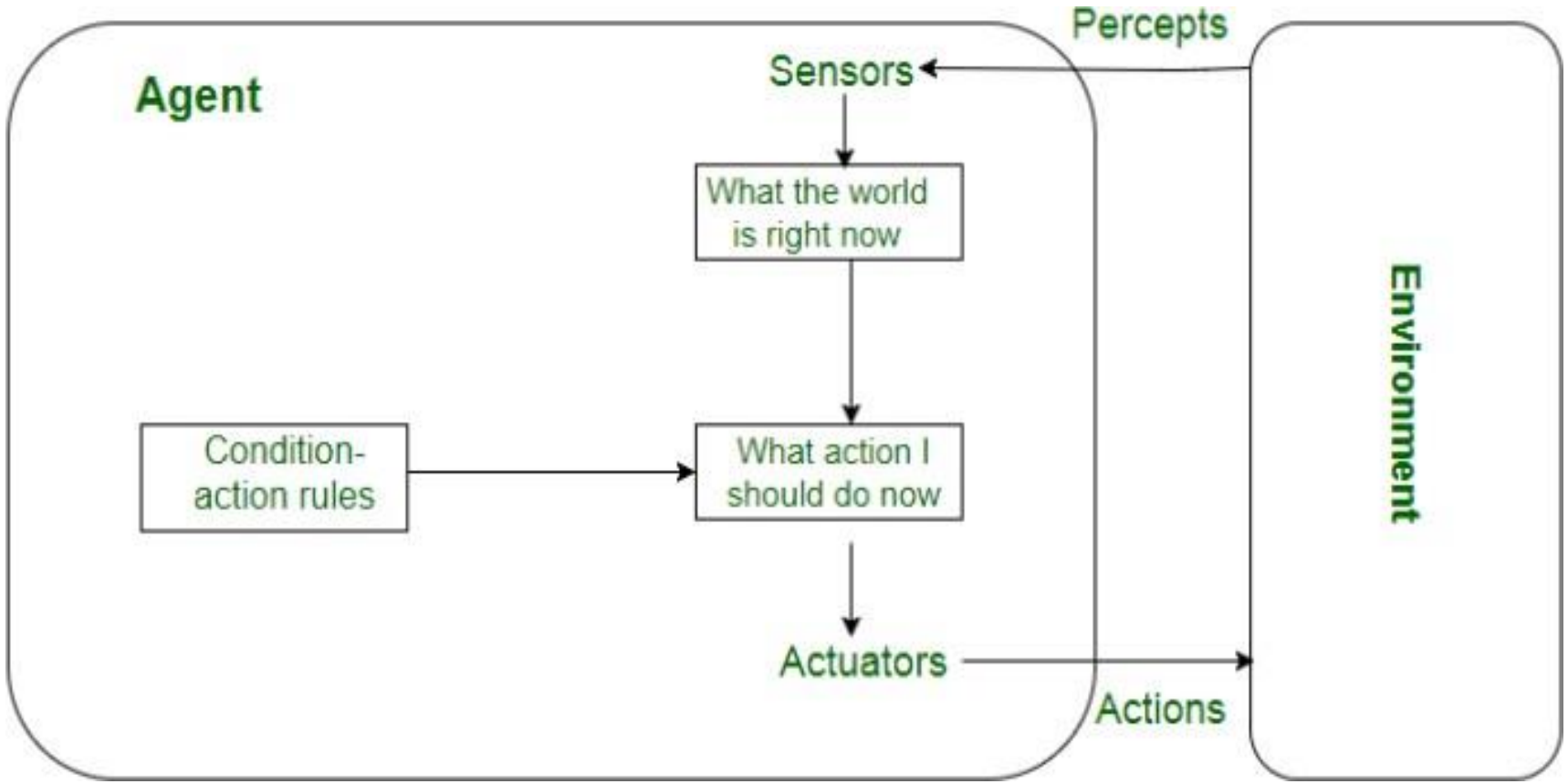


Example: the vacuum agent whose agent function is tabulated earlier is a simple reflex agent, because its decision is based only on the current location and on whether that location contains dirt.

Simple Reflex Agents

- Simple reflex behaviors occur even in more complex environments.
- Imagine yourself as the driver of the automated taxi. If the car in front brakes and its brake lights come on, then you should notice this and initiate braking.
- In other words, some processing is done on the visual input to establish the condition we call “The car in front is braking.” Then, this triggers some established connection in the agent program to the action “initiate braking.” We call such a connection a condition–action rule,
  - if *car-in-front-is-braking* then *initiate-braking*

Simple Reflex Agents



Simple Reflex Agents

- **The agent function is based on the condition-action rule.**
  - A condition-action rule is a rule that maps a state i.e, condition to an action. If the condition is true, then the action is taken, else not.
- **This agent function only succeeds when the environment is fully observable.**
- **Problems with Simple reflex agents are :**
  - Very limited intelligence.
  - If there occurs any change in the environment, then the collection of rules need to be updated.

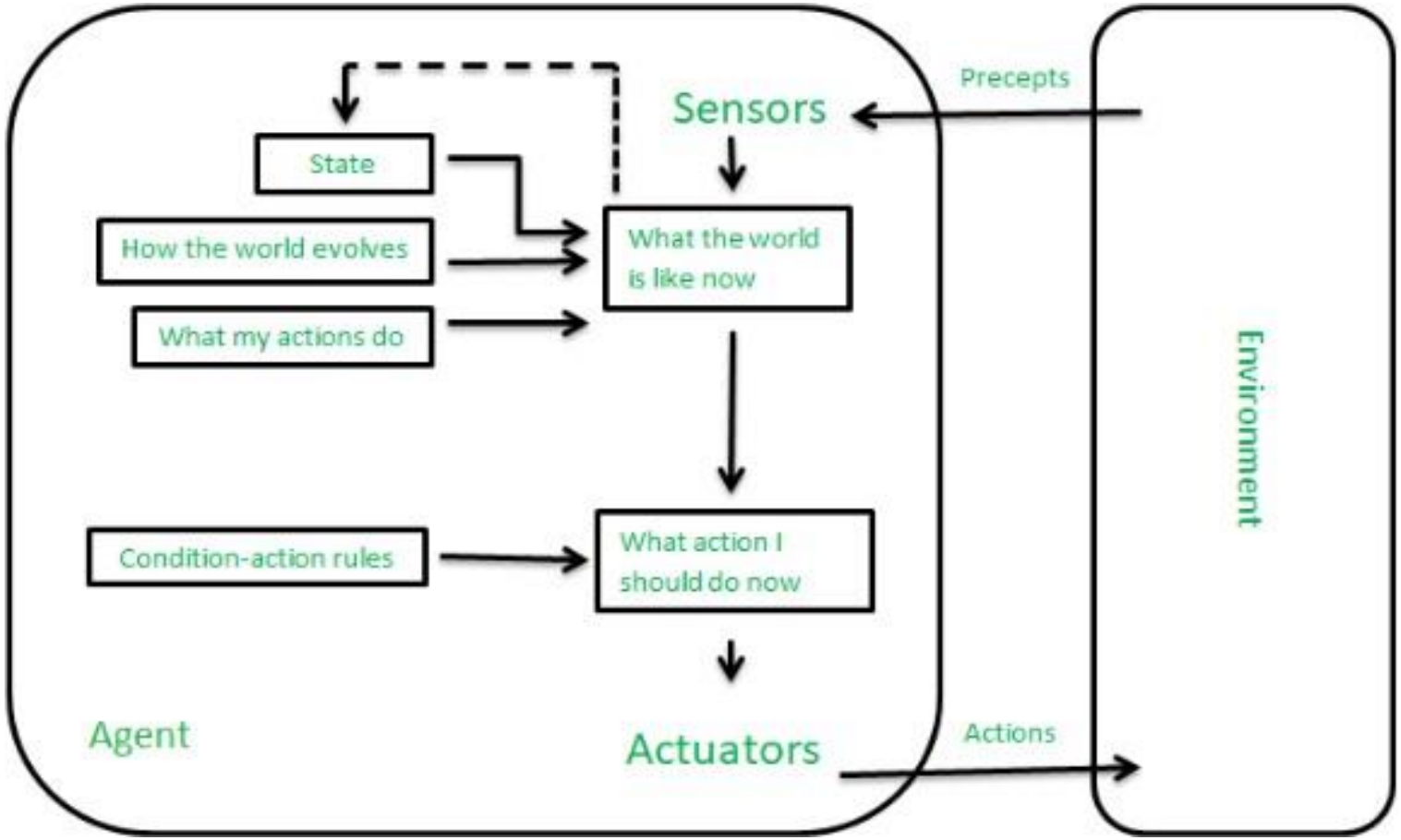
Model-Based Reflex Agents

- The most effective way to handle partial observability is for the agent to keep track of the part of the world it can't see now.
- That is, the agent should maintain some sort of *internal state* that depends on the percept history and thereby reflects at least some of the unobserved aspects of the current state.
    - For the braking problem, the internal state is not too extensive— just the previous frame from the camera, allowing the agent to detect when two red lights at the edge of the vehicle go on or off simultaneously.
    - For other driving tasks such as changing lanes, the agent needs to keep track of where the other cars are if it can't see them all at once.
    - For any driving to be possible at all, the agent needs to keep track of where its keys are.

Model-Based Reflex Agents

- Updating this internal state information as time goes by requires two kinds of knowledge to be encoded in the agent program.
  - First, we need some information about how the world evolves independently of the agent—for example, that an overtaking car generally will be closer behind than it was a moment ago.
  - Second, we need some information about how the agent's own actions affect the world—for example, that when the agent turns the steering wheel clockwise, the car turns to the right, or that after driving for five minutes northbound on the freeway, one is usually about five miles north of where one was five minutes ago.
- This knowledge about "how the world works" — is called a model of the world. An agent that uses such a model is called a ***model-based agent***.
- A model-based agent can handle partially observable environments by the use of a model about the world.

Model-Based Reflex Agents



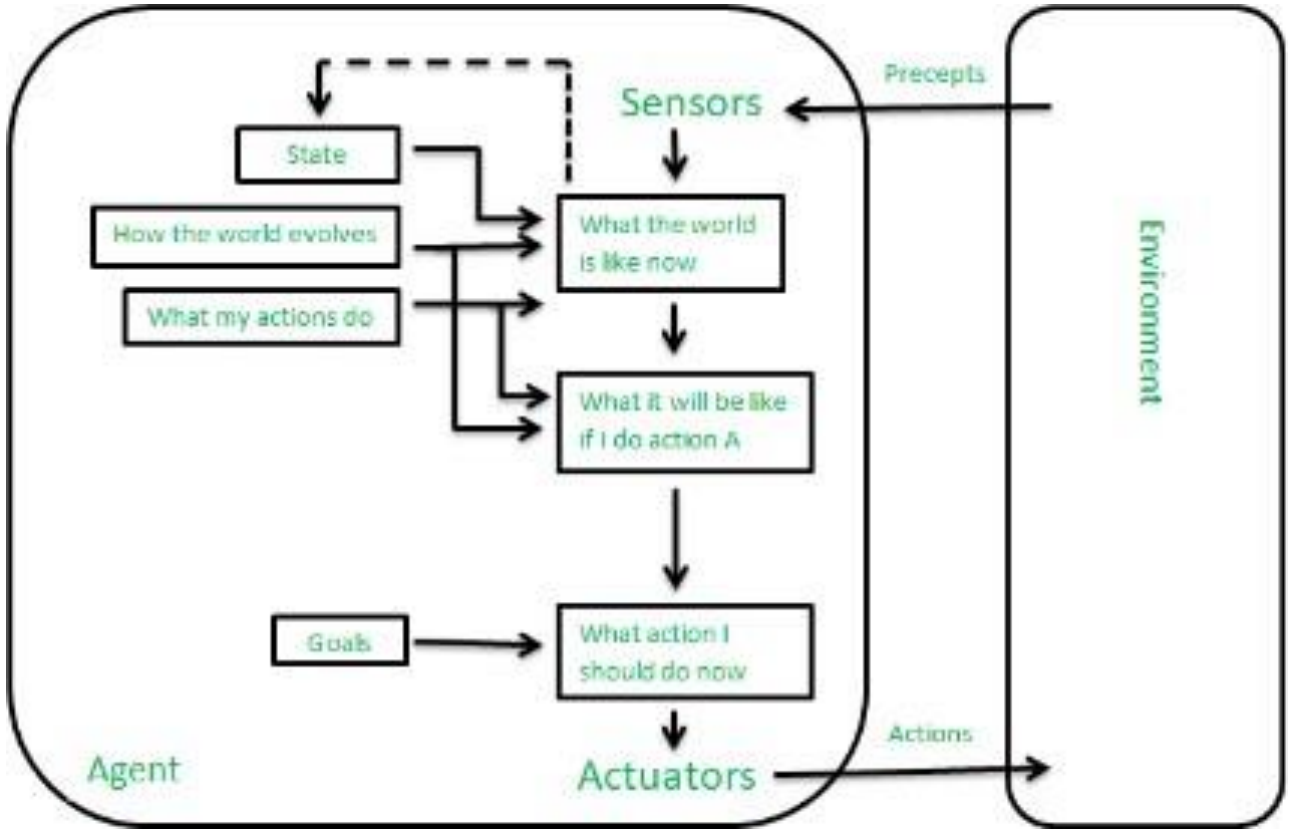
Goal-Based Agents

- Knowing something about the current state of the environment is not always enough to decide what to do.
- For example, at a road junction, the taxi can turn left, turn right, or go straight on. The correct decision depends on where the taxi is trying to get to.
- In other words, as well as a current state description, the agent needs some sort of goal information that describes situations that are desirable—for example, being at the passenger's destination.
- The agent program can combine this with the model (the same information as was used in the model based reflex agent) to choose actions that achieve the goal.

Goal-Based Agents

- These kinds of agents take decisions based on how far they are currently from their goal (description of desirable situations).
- Their every action is intended to reduce its distance from the goal.
- This allows the agent a way to choose among multiple possibilities, selecting the one which reaches a goal state.
- ***Search*** is one of the subfields of AI devoted to finding action sequences that achieve the agent's goals.

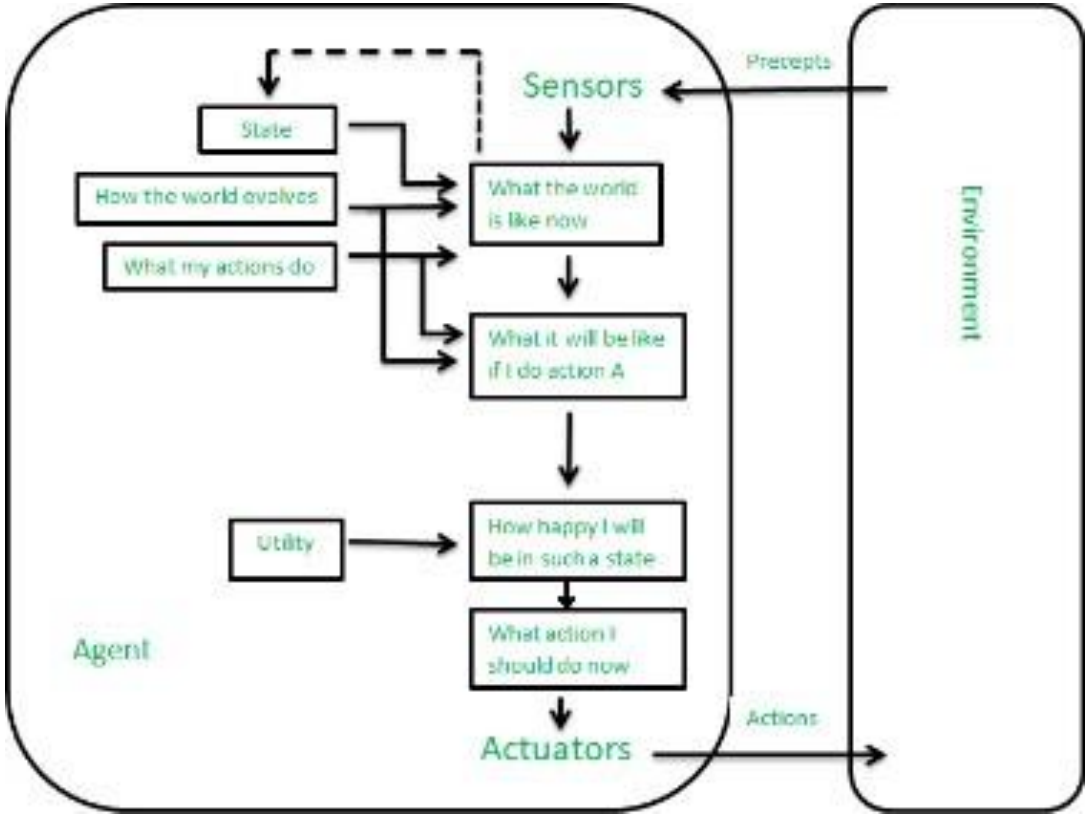
Goal-Based Agents



Utility-Based Agents

- Goals alone are not enough to generate high-quality behavior in most environments.
  - For example, many action sequences will get the taxi to its destination (thereby achieving the goal) but some are quicker, safer, more reliable, or cheaper than others.
- A more general performance measure should allow a comparison of different world states according to exactly 'how happy' they would make the agent, instead of only judging on the basis of 'happy' or 'unhappy' (goal-based approach).
  - Because 'happy' does not sound very scientific, economists and computer scientists use the term **utility** instead. Utility describes how 'happy' the agent is.
- When there are multiple possible alternatives, then to decide which one is best, *utility-based agents* are used. They choose actions based on a preference (utility) for each state.
- Sometimes achieving the desired goal is not enough. We may look for a quicker, safer, cheaper trip to reach a destination. Agent happiness should be taken into consideration. Because of the uncertainty in the world, a utility agent chooses the action that maximizes the **expected utility**.

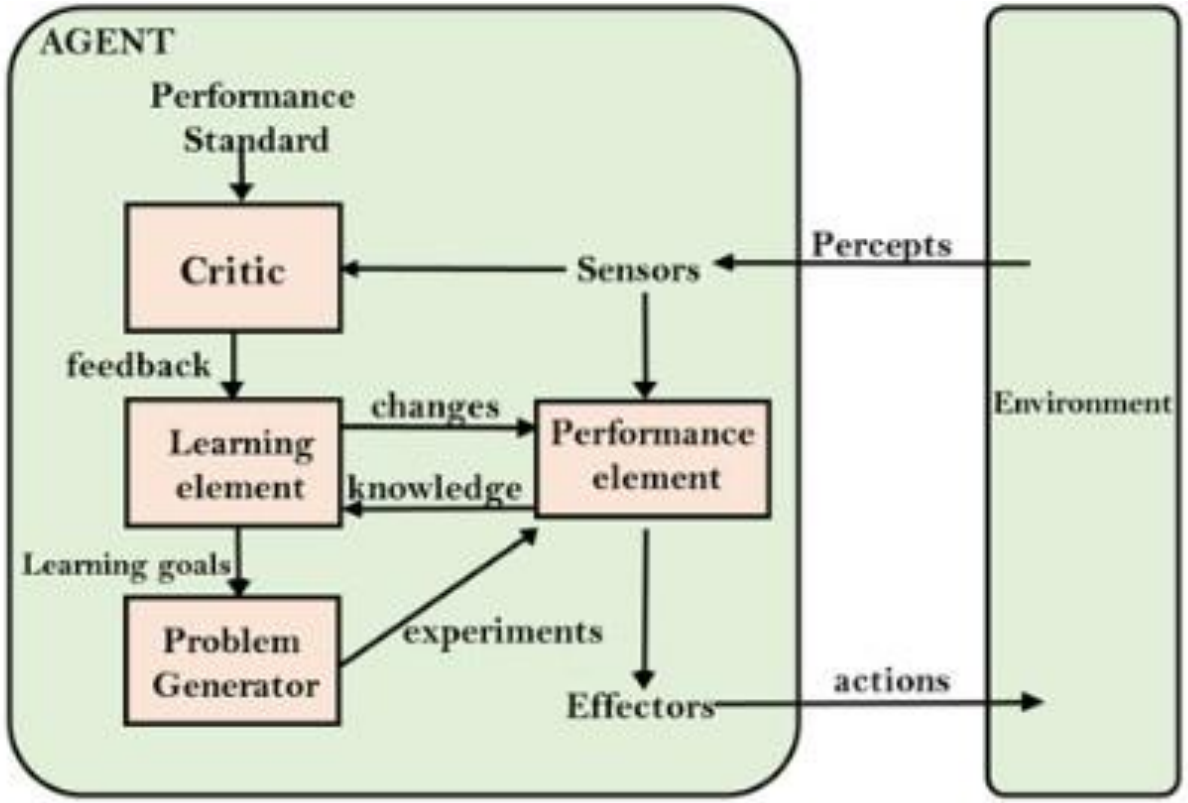
Utility-Based Agents



Learning Agents

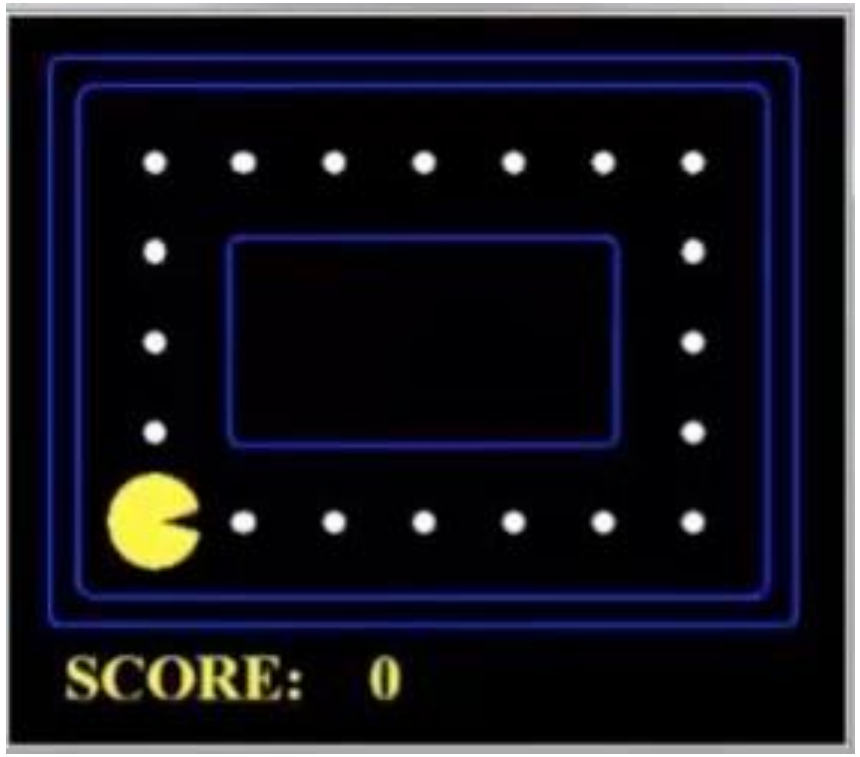
- A learning agent in AI is the type of agent that can learn from its past experiences or it has learning capabilities.
- It starts to act with basic knowledge and then is able to act and adapt automatically through learning.
- A learning agent has mainly four conceptual components, which are:
  - *Learning element:* It is responsible for making improvements by learning from the environment
  - *Critic:* The learning element takes feedback from critics which describes how well the agent is doing with respect to a fixed performance standard.
  - *Performance element:* It is responsible for selecting external action
  - *Problem Generator:* This component is responsible for suggesting actions that will lead to new and informative experiences.

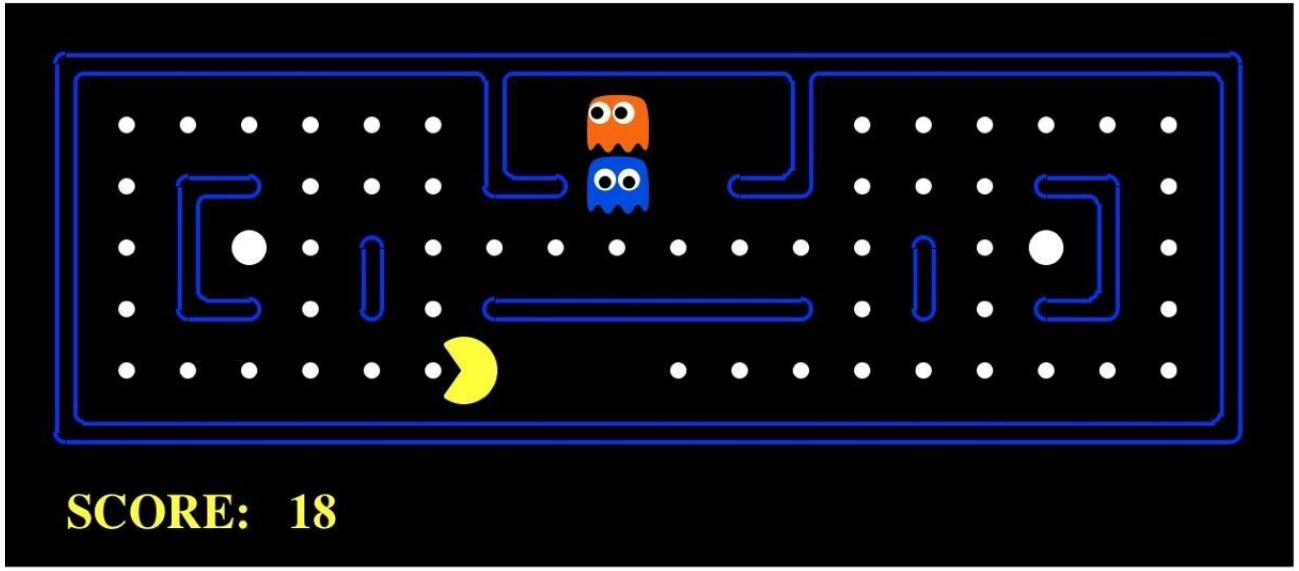
Learning Agents

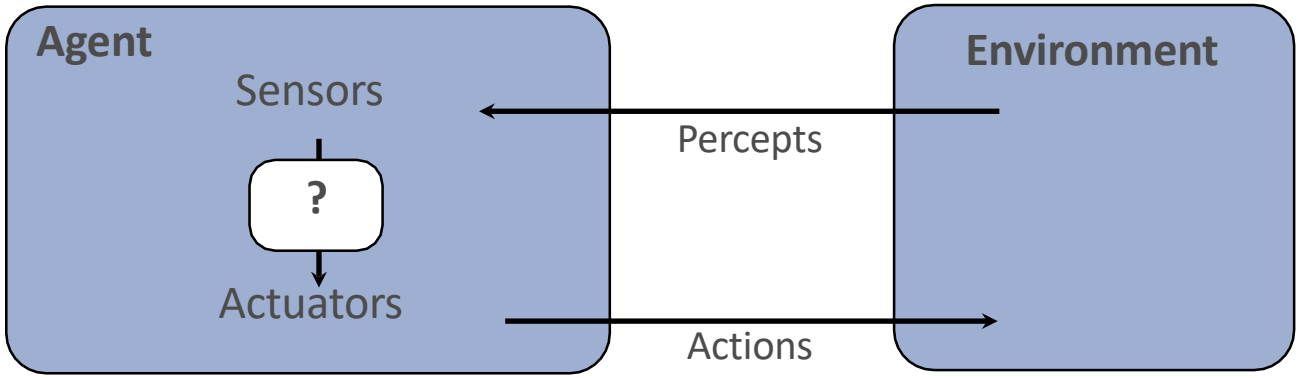


Learning in intelligent agents can be summarized as a process of modification of each component of the agent to bring the components into closer agreement with the available feedback information, thereby improving the overall performance of the agent.

Pac-Man as an Agent









Thank you for your attention!

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